RAII stands for "Resource Acquisition Is Initialization" What it means ? - A resource is acquired in a constructor and released in the destructor. What is a "resource" ? Anything that must be released: - memory (new) - file - lock - socket How RAII works ? 1. Object is created - constructor runs - resource is acquired 2. Object goes out of scope - destructor runs automatically - resource is released No manual cleanup required. RAII with smart pointers 1. unique_ptr - Constructor - owns memory - Destructor - deletes memory 2. shared_ptr - Constructor - increments count - Destructor → decrements count - Deletes memory when count = 0 Both follow RAII. RAII is a C++ technique where resource management is tied to object lifetime using constructors and destructors.